

High-dimensional data analysis of a wholesale distributor

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June 2024

Abstract

In our project we analyse the dataset *Wholesale customers data*, which contains information about the annual spending in monetary units (m.u.) undertaken by clients of a wholesale distributor on diverse product categories. To better understand the demand faced by the wholesale distributor, we investigate the available data to identify the most important spending features for different groups of clients. Moreover, we focus on the differences characterizing the underlying business when comparing different regions or different distribution channels. Therefore, we perform the Principal Component Analysis, which reduces the dimensionality of our dataset without losing relevant information. In particular we believe it is more efficient to map each client's demand by leveraging a smaller number of variables that capture the relevant features of the overall spending, rather than tracking the spending for each product category. Indeed, as we will show, PCA allows us to capture the dimensions along which the majority of the differences between and within channels occur. By contrasts, the spending levels across different regions seems to be quite homogeneous for each product category. Finally, leveraging the results of the PCA, we adopt clustering methods to better define groups of clients with similar features. The creation of new groups of clients characterized by similar features can represent a first step to introduce new segments of the business in addition to the existing ones (*Channel* and *Region*).

1 Descriptive statistics

The dataset *Wholesale customers data* has a total of 440 observations for 8 different variables, and does not contain any missing value. The variables included in our dataset are the following ones:

- **Channel:** customers' Channel - Horeca (Hotel/Restaurant/Café) (1) or Retail channel (2);
- **Region:** customers' Region - Lisbon (1), Oporto (2) or Other (3);
- **Fresh:** annual spending (m.u.) on fresh products;
- **Milk:** annual spending (m.u.) on milk products;
- **Grocery:** annual spending (m.u.) on grocery products;
- **Frozen:** annual spending (m.u.) on frozen products;
- **Detergents paper:** annual spending (m.u.) on detergents and paper products;
- **Delicatessen:** annual spending (m.u.) on delicatessen products;

As we can see in the previous list, our dataset contains two categorical variables, namely *Channel* and *Region*, and six quantitative ones. The variable *Region* refers to the three geographical areas where the observations come from (*Lisbon*, *Oporto* or *Other*). The variable *Channel* represents the distribution channel where the expenses were recorded, which can be either Hotel/Restaurant/Cafe(*Horeca*) or *Retail*. Specifically, in our dataset, we have 298 observations belonging to *Horeca* category and 142 belonging to *Retail* category. In contrast, when we look at the geographical origin of our observations we see that the region with the largest number of observations is

	Mean	Sd	Min	1st Qu	Median	3rd Qu	Max
Fresh	12000.0	12647.3	3.0	3128.0	8504	16934.0	112151.0
Milk	5796.0	7380.4	55.0	1533.0	3627	7190.0	73498.0
Grocery	7951.0	9503.2	3.0	2153.0	4775.5	10656.0	92780.0
Frozen	3071.9	4854.7	25.0	742.2	1526	3554.2	60869.0
Detergents paper	2881.5	4767.9	3.0	256.8	816.5	3922.0	40827.0
Delicatessen	1524.9	2820.1	3.0	408.2	965.5	1820.2	47943.0

Table 1: Summary statistics

"Other", where as *Lisbon* and *Oporto* report respectively 77 and 47 observations. Once we have reviewed the general features of the categorical variables included in our dataset, we can focus on the other 6 quantitative variables. The statistical features of these variables are summarized in **Table 1**.

We can immediately notice that the variables are heterogeneous in mean, standard deviation and range of expenditure. Considering the mean we pass from a minimum of 1524.9 m.u. for *Delicatessen* to 12000 m.u. for *Fresh*. Considering standard deviation, again, the minimum refers to *Delicatessen* (2820.1 m.u.) and the maximum to *Fresh* (12647.3 m.u.). As regards the range, the one of *Fresh* is by far greater (more than the double) than the one of *Detergent Papers* or of *Delicatessen*. Indeed, as we can see from the **Table 1**, for each variable the mean is larger than the median and the maximum value tends to be extraordinary larger than both the mean or the median and more than one standard-deviation distant from the 75th percentile. These values suggest us that our variables have a right-skewed distribution with many outliers on the right tail.

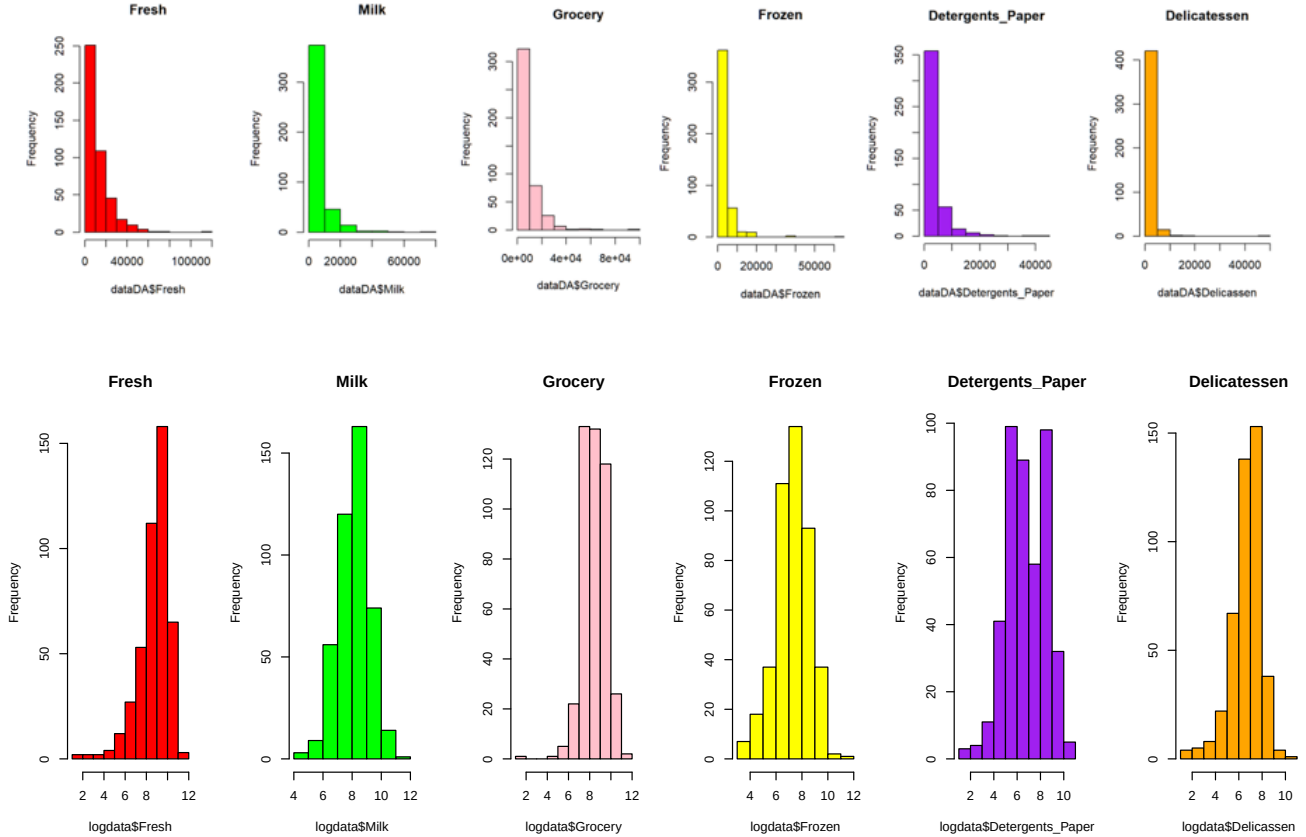


Figure 1: Histograms showing variable's distribution before and after log transformation and standardization

As we can see from the histograms above (**Figure 1**), we graphically confirm the idea about positively skewed distributions. To deal with this feature of our data, we then perform a logarithmic transformation of these six

quantitative variables. We can apply it because our variables are strictly positive and continuous. This kind of transformation allows us to reduce the skewness of the distributions, which become closer to the normal one, and the weight of the outliers characterizing our variables of interest.¹

Moreover, as we are interested in highlighting the differences between the groups of clients represented by each channel and each region, we believe it is worthwhile to analyse how the distribution of our six spending categories changes when we control for either the *Channel* or the *Region*. In order to do so, we report the box plots for the expenditure level in each product category divided by *Channel* and *Region*.

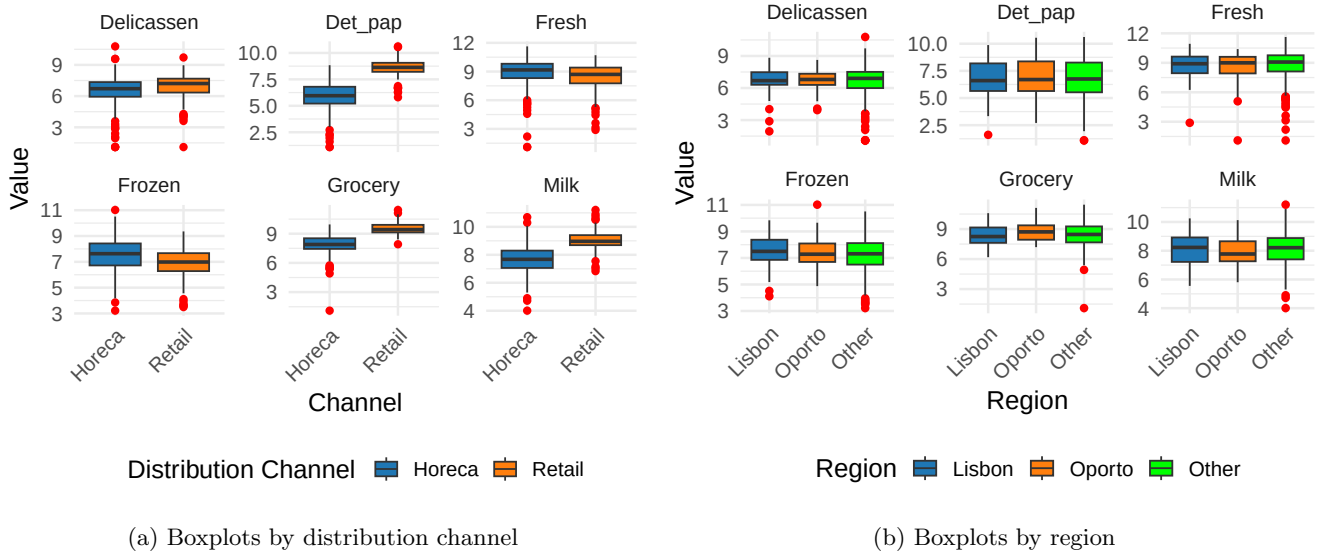


Figure 2: Boxplots

As it is shown in the **Figure 2**, we can see that, as expected, the *Retail* channel reports higher average values for *Grocery*, *Milk* and *Detergents Papers*. In contrast, the *Horeca* channel reports slightly higher spending levels for frozen and fresh product. This confirms our original expectations about the differences between the two lines of business. However, when we control for the geographical location, we do not find any important differences in the spending level for one of the products included in our sample. Therefore, we can conclude that the demand faced by the wholesale is on average very homogeneous when we compare the three regions of interest, where as it seems to present very remarked differences when we compare the two distribution channels.

Finally, as we want to analyse the relationship between the spending for these six product categories, we look at the correlation between them. Given the different values of the variance among our variables of interest, it is better to look at the correlation matrix rather than the variance-covariance one. This same concept will be applied subsequently when performing the PCA. The correlation matrix for these variable is reported in **Table 2**.

We can observe that some variables display some high positive correlation between each other: *Grocery* - *Detergent paper* (0.80), *Grocery* - *Milk* (0.76) and *Milk* - *Detergent paper* (0.68). At the same time *Fresh*, *Frozen* and *Delicatessen* display relevant positive correlations ranging from 0.25 to 0.38. Moreover, the expenditure level in *Milk* and *Delicatessen* is quite correlated, as the two variables show a correlation of 0.34. Negative correlation among variables are also present, but they are very low in magnitude: *Fresh* - *Milk* (-0.02), *Fresh* - *Detergent paper* (-0.16), *Milk* - *Frozen* (-0.06), *Frozen* - *Detergent paper* (-0.21). The two groups of correlated variables (*Milk*-*Grocery*-*Detergent papers* and *Fresh*-*Frozen*-*Delicatessen*) can be considered as an interesting guide for the following PCA and its possible results.

¹The following analysis are performed on the logarithmic transformation of our data

	Fresh	Milk	Grocery	Frozen	Detergent paper	Delicatessen
Fresh	1.00	-0.02	-0.13	0.38	-0.16	0.26
Milk	-0.02	1.00	0.76	-0.06	0.68	0.34
Grocery	-0.13	0.76	1.00	-0.16	0.80	0.24
Frozen	0.38	-0.06	-0.16	1.00	-0.21	0.25
Detergent paper	-0.16	0.68	0.80	-0.21	1.00	0.17
Delicatessen	0.26	0.34	0.24	0.25	0.17	1.00

Table 2: Correlation matrix

2 Principal component analysis

2.1 Theory

The implementation of the PCA is justified by the presence of highly correlated variables in the dataset. The aim of the analysis is to simplify our data structure by reducing a large number of highly correlated variables to a few independent principal components, losing as little information as possible. Each principal component W_k is an unobservable linear combination of the variables ($W_k = \sum_{j=1}^d \eta_{kj} x_j, k = 1, \dots, d$) reflecting the direction of highest variance in the data and satisfying the following three assumptions:

- they have to be uncorrelated: $\text{cov}(W_k, W_{k'}) = 0$
- they have to be ordered in decreasing order according to their variability (so that the first principal component is the best one-dimensional summary of the data): $V(W_1) \geq \dots \geq V(W_k) \geq \dots \geq V(W_d)$
- the total variance of the two systems have to be the same: $\sum_{k=1}^d V(W_k) = \sum_{j=1}^d V(X_j)$

Therefore, given our initial set of d variables, to get $k \leq d$ components, we start from the first component W_1 . Defining the d dimensional subspace spanned by $(x_1, \dots, x_d)$, the line OW_1 is determined such that it maximizes the dispersion of its projection on the plane. Once we have found OW_1 , to determine the second component, we consider the $d - 1$ dimensional space orthogonal to OW_1 . OW_2 is the line that maximizes the dispersion of its projection in this new plane.² We can go on with this procedure until defining as many components as the number of variables. It follows, hence, that each principal component W_k is a linear combination of the variables $(x_1, \dots, x_d)$, such that $\sum_{j=1}^d \eta_{kj}^2 = 1$ (components are orthogonal) and $\text{Var}(W_k)$ is maximal.

Once we have computed all the components, we have to decide the number of them that is useful to keep for our purposes. In order to select the best number of components, we can rely on different criteria, which we apply in our analysis. Each of these criteria aims at minimizing the loss of information by retaining the smallest number of components possible. η_{kj} is the weight exerted by the variable x_j on the principal component W_k and it is called loading. However, when the correlation matrix is analysed, the standardized loadings ($\eta_{kj}^* = \sqrt{\lambda_k} \eta_{kj}$) are particular useful for the interpretation of the components themselves. In fact, by interpreting the selected components, we can relate them to the original variables and better understand the statistical information underlying our original dataset, which was more difficult to interpret given the larger number of variables.

2.2 Analysis

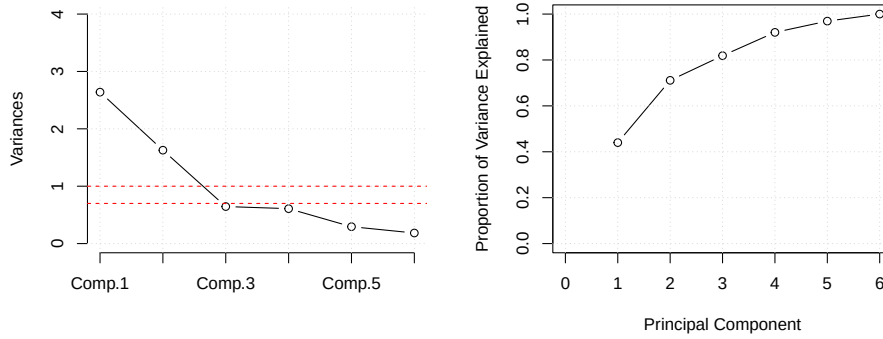
We want to use PCA in order to reduce the number of variables which characterize a client's expenditure. Given the differences in the variances of our variables of interest, we decided to perform the PCA on the correlation matrix. In fact, if we used the variance-covariance matrix, rather than the correlation one, we would give more weight to those variables showing a greater variance. Therefore, we deal with the different range of variability of our variables by using the correlation matrix, which allows us to give all the variables the same weight regardless the magnitude of their own variance. The variances of the principal components are represented by the eigenvalues of the correlation matrix, while the loadings are represented by the element of the associated eigenvectors. **Table 3** below summarizes the results obtained:

²Iterating these steps, we have that the k -th component, for example, lies on the $d-k+1$ dimensional subspace orthogonal to OW_{k-1} .

	PC1	PC2	PC3	PC4	PC5	PC6
Standard deviation($\sqrt{\lambda_k}$)	1.625	1.276	0.803	0.78	0.543	0.429
Proportion of Variance	44%	27.1%	10.8%	10.1%	4.9%	3.1%
Cumulative Proportion	44%	71.1%	81.9%	92%	96.9%	100%

Table 3: PCA

By looking at the last row, which displays the cumulative proportion, we notice that the first two components explain 71.1% of the total variability in the dataset, which rises to 82% if we consider also the third component. The graphical representation is reported in the right panel of **Graph 2** and it suggests that we can consider two or three components for our analysis. In fact, one possible way to select the exact number of components is by looking for the "elbow" in the scree plot (left panel of Graph 2). In this case we identify the "elbow" for the third component. However, we think that keeping also the third component can be misleading as it shows a variance (λ_3) equal to 0.65 and significantly lower than the one of the first two components (respectively $\lambda_1 = 2.64$ and $\lambda_2 = 1.63$). In addition to this large decrease in the variance captured by the third component, the rule of thumb is to retain only those components whose eigenvalues is greater than 1 or 0.7 (in the scree plot in the left panel of Graph 2 the both thresholds are reported with a red, dotted line). Therefore, as the variance of third component is equal to 0.65, the previously defined condition is not satisfied. Taking into consideration these results, we decided to select only the first two components.



Graph 2: Eigenvalues (*left*) and cumulative distribution to variance in % terms (*right*) against the number of PCs

At this point we proceed by computing the standardized component loadings and we report the ones associated to the first two in **Table 4**³. As we are analysing the correlation matrix, the loadings η_{kj}^* are interpreted as the correlation coefficient between variable j and component k . The relative sizes of the η_{kj}^* 's reflect the relative contributions made by each variable to the component.

	η_{k1}^*	η_{k2}^*
Fresh	0.17	0.75
Milk	-0.88	0.17
Grocery	-0.93	-0.01
Frozen	0.22	0.75
Detergents Paper	-0.90	-0.09
Delicatessen	-0.34	0.68

Table 4: Loadings for the first two components

³For the third component we get standardized loadings difficult to interpret. The lack of interpretation for the third component seems to support our idea that it does not carry relevant information for our analysis

We observe that the first component has high and negative loadings for *Milk* ($\eta_{12}^* = -0.88$), *Grocery* ($\eta_{13}^* = -0.93$) and *Detergent Paper* ($\eta_{15}^* = -0.90$), while the second component has high and positive loadings for *Fresh* ($\eta_{21}^* = 0.75$), *Frozen* ($\eta_{24}^* = 0.75$) and *Delicatessen* ($\eta_{26}^* = 0.68$). Therefore, the first component can be interpreted as the spending on common goods, while the second as the expenditure on more specific kind of food like fresh and frozen food. In particular, the first component tends to assume lower values for clients who spend more on those three products (*Milk*, *Detergent papers* and *Grocery*), whereas the higher is the expenditure on *Fresh*, *Frozen* and *Delicatessen* the higher is the value of the second component. Thanks to our PCA we are able to differentiate members of different distribution channels mostly by means of the first principal component. In fact, this component is negatively and highly correlated with those variables (*Milk*, *Grocery* and *Detergent Papers*) which report greater differences between the two distribution channels. Lower values of the first component are associated with observations belonging to the retail channel where the spending on those common goods is on average higher. We can appreciate this distinction in the graph reported below (**Figure 3**). At the same time, the second component, allows us to distinguish clients with higher spending on *Fresh*, *Frozen* and *Delicatessen* from the ones characterized by a lower level of spending on these three products. As we can see in (**Figure 3**), the variance of the spending on those three products (*Fresh*, *Frozen* and *Delicatessen*) is not related to the differences between channels but to the differences within each channel category. In fact, the second component can assume either high or low values equally for both *Horeca* and *Retail*. Again we can appreciate this detail in the figure reported below. By contrast, we cannot use the two components to distinguish the observations belonging to different regions. In fact, as already discussed, the distribution of the spending level for these products seems to be homogeneous across the three regions.

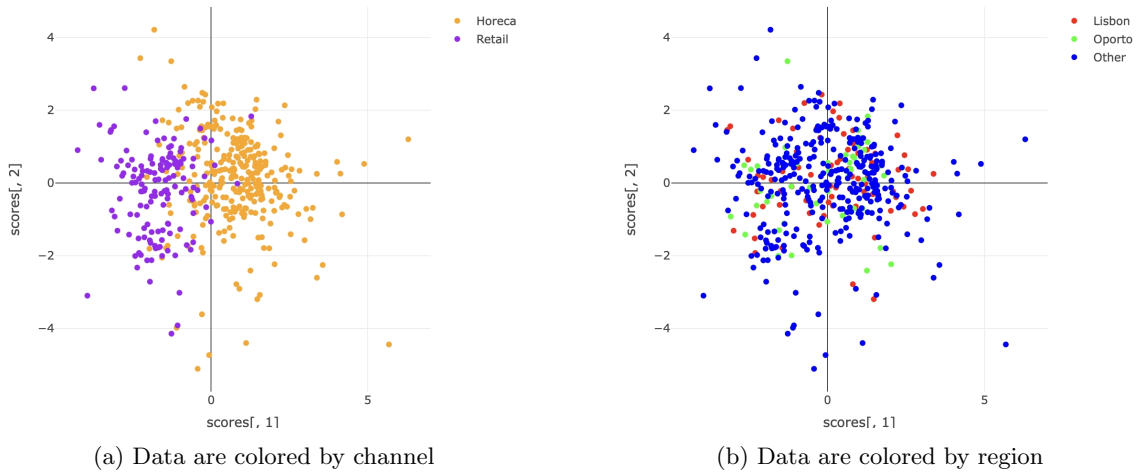


Figure 3: Scores for first and second components

In conclusion, it is interesting to notice that observations belonging to different channels can be broadly individuated by the means of the first component. Indeed, the first component reflects the differences in the spending for *Milk*, *Grocery* and *Detergent papers* which are associated with the differences between the two channels (variance between). In contrast, the second component captures the variance mostly associated with the differences in the spending for *Fresh*, *Frozen* and *Delicatessen*, which, as shown in the **Figure 2** and in the **Figure 3** seem to explain differences within channels.

3 Cluster analysis

3.1 Theory

In cluster analysis, number and identification of clusters is not known *a priori*. Therefore, the method works in two steps: (i) measuring distance between observations; (ii) developing a *method of clustering* (i.e. building clusters based on distances).⁴

⁴Notice that, given that our sample only contains continuous variables, this brief theoretical explanation of the method will not cover categorical variables techniques.

As for (i), given a $n \times d$ matrix of observations X , distances (between two generic observations i and j) are computed using the *Minkowski distance*:

$$\delta_{ij} = \left\{ \sum_{k=1}^d |x_{ik} - x_{jk}|^l \right\}^{\frac{1}{l}} \quad (1)$$

where l is a tuning parameter that determines the weight given to large distances (the higher l , the larger the weight).

Once done this, we need to determine how many clusters to consider and how observations are allocated among them (ii). There are two main categories of clustering methods: *hierarchical* and *non-hierarchical*. Hereby, we will list the methods we will use in this analysis. Except for the last one, they are all *hierarchical*. Looking at agglomerative methods only, we start from n clusters (i.e. one per observation) and aggregate them in a step-by-step procedure.

Nearest neighbor: we look at the $n \times n$ distance matrix D , and in particular at the generic entry δ_{ij} (computed as in (1)) with the smallest value. We aggregate cluster i and j into *cluster A* (C_A). We proceed by looking at the new $(n-1) \times (n-1)$ distance matrix where the distance between $C_A = \{i, j\}$ and *cluster B* $\in n \setminus \{i, j\}$ is $\delta_{C_A C_B} = \min\{\delta_{iB}, \delta_{jB}\}$. The method proceeds like this until one reaches the desired level of aggregation.

Farthest Neighbor: it works exactly like the Nearest neighbor method, just setting $\delta_{C_A C_B} = \max\{\delta_{iB}, \delta_{jB}\}$ (i.e. taking the highest distance between the objects inside a cluster). Notice that as n increases, it is more likely that results from the two methods differ substantially. Instead, if they are similar, it means that the allocation is robust to different methods.

Centroid: the two above methods tend to over/under-estimate the differences between observations, therefore we can build clusters based on the distance between the center of each one (i.e. the average of observations inside each cluster). Hence, the distance between C_A and C_B is $\delta_{C_A C_B} = \sqrt{\sum_{k=1}^d (\bar{x}_{kA} - \bar{x}_{kB})^2}$.

Ward method: at each step of clustering procedure we check how much information we lose by pairing observation i and j . Therefore, i and j are chosen such that to minimize this information lost as measured by the MSE $W_s = \sum_{k=1}^d \sum_{i=1}^{n_k} (x_{iks} - \bar{x}_{ks})^2$ (where the first sum captures the information lost for variable k at the s^{th} clustering step.)

K-means clustering: this method assumes k clusters *a priori* and allocates each observation into one of them. The objective is to minimize variability within each cluster. We start by selecting k observations acting as centroid of each cluster and allocating the $n-k$ remaining ones into the k clusters. This first partition is usually improved by using the actual centroid of every cluster as new aggregation point.

3.2 Analysis

We run the analysis with both hierarchical and non-hierarchical clustering techniques.⁵

By the means of this analysis we want the model to find on its own an underlying structure for the data. From a theoretical point of view, good clusters are such that they have high BSS (Between sum of squares) among each other and low WSS (Within sum of squares). In other words, they are well separated from each other but compact around their own center. In order to have a more objective way of assessing cluster's quality we will rely on the Silhouette score; this measure ranges from -1 to 1 and it is closer to 1 the more observation matches its own cluster, differently it is close to 0 the more there is uncertainty around the classification and it is negative if it should have probably assigned to a different cluster.

By looking at the hierarchical dendograms (see **Figure 4**), the Ward method is the only one that seems to be an eligible candidate for our analysis; in particular we discard methods that tend to create clusters with very few observations. By looking at the Ward's dendogram we can notice that two clusters seem to be a good cut. We can perceive from the plot the two clusters seem to contain observations that are quite close within themselves but meanwhile the two clusters "need" a significant distance two group together, this is a signal of good separation between them. Afterwards we compute Non-hierarchical method k-means with k from 2 to 5. In this case we can use WSS/TSS ratio as one of the tool to assess the optimal number of clusters. This ratio decreases in k by construction with the number of cluster and so we have to face a trade-off. By the analysis of the latter, we notice that it quickly decrease in k already after the second cluster. Another tool we can rely on is the average silhouette score which is higher for $k = 2$ (see **Table 5**). If we consider those results the optimal number of cluster seems to

⁵The following analysis are performed on the standardized logarithmic transformation of our data.

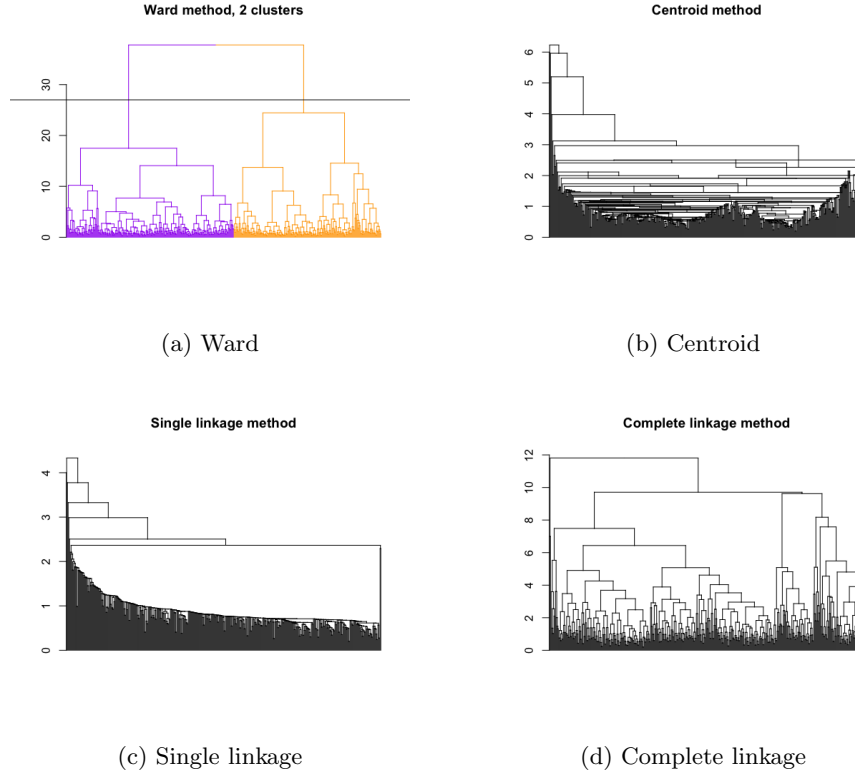


Figure 4: Dendrograms of the hierarchical methods

be two, this is even strengthened by the fact that by manually checking the Ward gives exactly the same clusters of the k-means (2 clusters case). In addition, the k-means seem to perform better in terms of silhouette (see **Table 5**) and does not show any kind of overlapping (see **Figure 6.a**). Thus it seems to represents the best choice in order to analyze our data.

A two cluster separation is consistent with what we see from plotting the two first components with observations colored by channel: clients tend to differentiate depending on how much they spend in goods like milk, grocery and detergents/paper. As a matter of facts it turns out that the two clusters resemble (even if not perfectly, we highlight a tendency not a rule) to be the two kinds of Channel . Clients that has an higher expense on milk, grocery and detergents/paper tend to belong to retail shops category (orange cluster), while consumers that tend to buy less those kind of goods are more incline to belong to Horeca (purple cluster).

As a final consideration, we notice that exploiting the variability carried by the second component, we can differentiate consumers within the channel group. In other words we previously found that consumers can be divided by channels, but the second component can ideally be used to divide consumers belonging to the same channel. Hence one can think to use a k-means model with $k = 4$, but we saw that the latter has a poor average silhouette score of 0.21 with a lot of negative values for some observations (see **Figure 5.c**), even if a taste of the aforementioned separation can be noticed in the data (see **Figure 6.b**).

Hence cluster analysis suggest an underlying structure of the data composed by 2 clusters, discarding other solutions that even if theoretically interesting do not find a strong confirmation in the data.

	2	3	4	5
Ward	0.26	0.23	0.19	0.18
K-means	0.29	0.26	0.21	0.2

Table 5: Average Silhouette scores for Ward and K-means methods

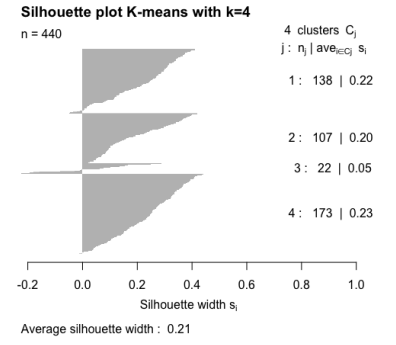
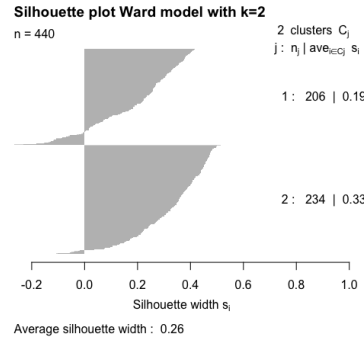
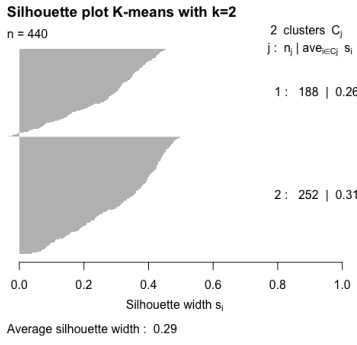
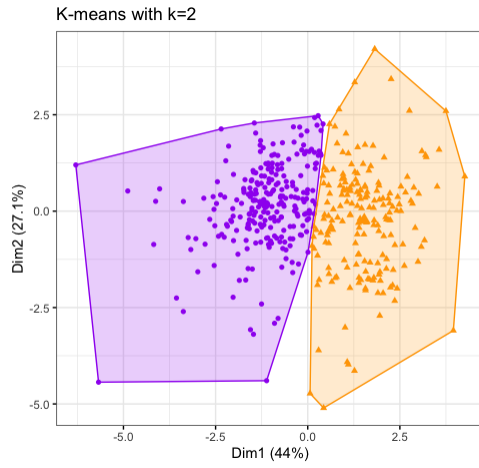
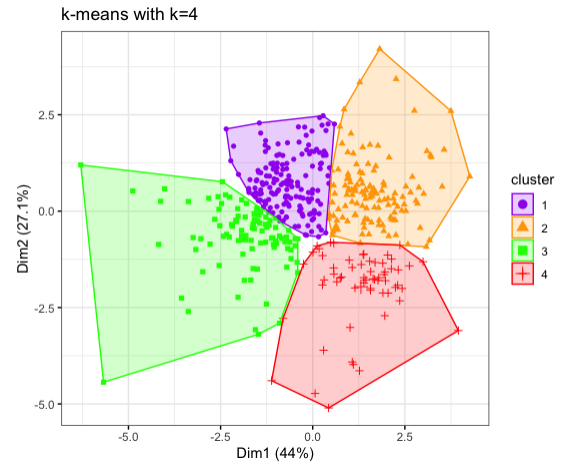


Figure 5: Silhouette scores



(a) 2 Cluster



(b) 4 Clusters

Figure 6:K-means clusters plotted on the first and second component

4 Conclusion

In this project we implemented the Principal Component Analysis and the Cluster Analysis as they were the most appropriate techniques for studying the *Wholesale customers* dataset. For each technique, we decided to provide both a theoretical framework, the practical application and the justification of its use to be as complete and rigorous as possible. From the PCA we were able to reduce the dimension of the dataset by selecting two principal components. Interestingly, we discovered that the first component explains the differences in observations between the two different channels (Horeca and Retail), while the second component captures the differences within channels. Moreover, by interpreting the standardized loadings we could identify the groups of products related to those differences. With the Cluster Analysis, instead, we deepened our study through the application of several clustering techniques. The results suggest the presence of an underlying structure which characterises the dataset. Specifically, the presence of two clusters strongly emerges.